Random diagonally stable gLV community with antibiotic exposure

Model equations and parameter meanings

For species i=1,...,n:

$dX_i/dt = X_i [r_i(1 - c\ rho_i) + \Sigma_{j=1}^{n} A_{ij}X_j - k_{max}A(t)^h/(IC50_i^h + A(t)^h)]$.

X_i is abundance; r_i is intrinsic growth; A_{ij} is the per-capita effect of species j on i; rho_i in [0,1] is resistance; c is the maximum fractional growth cost at rho=1; and the Hill term is direct antibiotic killing.

Antibiotic dynamics: $dA/dt = -k_{deg}A$; $A(0)=A_0$.
Resistance mapping: $IC50_i = IC50_{min}(IC50_{max}/IC50_{min})^{rho_i}$.

There is no explicit resource, beta-lactamase, or inhibitor state. The ecological interaction matrix is randomly sampled, then constructed as $A=W^{-1}(K-P)$, with W diagonal positive, K skew-symmetric, and P symmetric positive definite. Thus WA+A^T W = -2P is negative definite, certifying diagonal stability. A positive target equilibrium x* is sampled and r=-Ax*, so the no-drug, zero-cost system has a feasible equilibrium by construction.

What was run

Matched no-drug controls; antibiotic dose and decay sweeps; species-level resistance selection relative to matched control; resistance-cost tradeoff; representative growth-versus-killing trajectories; initial-biomass-by-dose and interaction-strength-by-dose factorials; a controlled resistance-to-species assignment experiment with fixed ecology and a fixed rho multiset; a 40-community rank-correlation sensitivity screen; and a multi-seed robustness check. Since gLV has no explicit resource pool, initial biomass and interaction strength replace CRM resource availability/leakage as context factors.

Stability and interpretation checks

max_eigenvalue_symmetric_part: -2

min_eigenvalue_P: 1

max_abs_diagonal_error: 1.11e-16

max_equilibrium_residual: 0

Scenario ranges

Parameter	Values / range	Interpretation
Community	n=25; interaction scale 0.25 baseline	Fixed community for matched sweeps; separate stable matrices for ensemble analyses
Drug dose A_0	0, 0.1, 0.25, 0.5, 1, 2	Initial antibiotic concentration (normalized units)
Drug decay kdeg	0, 0.005, 0.01, 0.02, 0.05, 0.1	First-order clearance; 0 means no clearance
Resistance cost c	0, 0.1, 0.2, 0.3, 0.4	Growth cost at resistance score ρ=1
IC_{50}	0.05–2.0; log-interpolated by ρ	Species-specific antibiotic susceptibility trait
Initial biomass	0.0025, 0.01, 0.04, 0.16	Density-dependent gLV context (CRM resource analog)
Sensitivity screen	40 draws; kmax 0.15–0.8; h 1–4	Exploratory Spearman screen; other ranges listed above

scenario	final_biomass	survivors	final_shannon	final_weighted_resistance
control	4.665	25	3.2144	0.5195
dose=1	4.2074	24	3.1264	0.5606
dose=2	3.0367	24	2.8281	0.6278
decay=0	0.0766	3	0.0001	0.8927
decay=0.1	4.665	25	3.2144	0.5195

Run-specific findings

In the fixed-seed dose sweep, final biomass changed from 4.66 without drug to 3.04 at A_0=2 (-34.9%). Survivors changed from 25 to 24, while biomass-weighted resistance changed from 0.520 to 0.628. This is consistent with preferential retention of initially more resistant community members; resistance is assigned at initialization and does not evolve here.

At A_0=1, the no-decay and fastest-decay cases ended at 0.0766 and 4.66 total biomass, respectively. This captures exposure-duration effects as well as peak dose.

The three largest absolute rank correlations with final biomass in the 40-draw sensitivity screen were final_shannon (|ρ|=0.86), survivors (|ρ|=0.73), final_weighted_resistance (|ρ|=0.65). They are exploratory associations across joint parameter/community draws, not isolated causal effects.

The untreated resistance-cost sweep changed final biomass from 5.2 at c=0 to 4.13 at c=0.4. This is the expected direct fitness-cost effect and is why antibiotic response should be interpreted against a matched control at each cost.

At A_0=1, initial-biomass scenarios spanned final biomass 4.14–4.35; interaction-strength scenarios spanned 2.43–3.25. The latter use separately sampled stable matrices, so this spread includes the associated change in feasible growth vector, not only interaction magnitude.

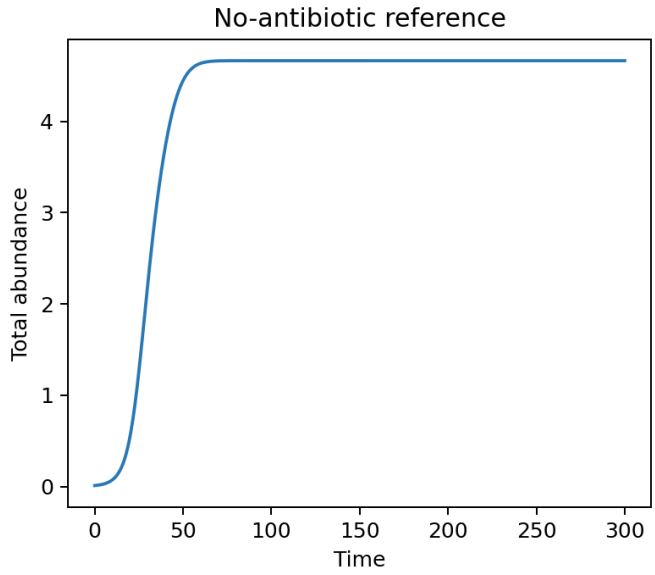
Across the ten independently sampled communities at A_0=1, final biomass median was 3.47 (range 2.71–3.99); these seeds vary both ecology and resistance assignment.

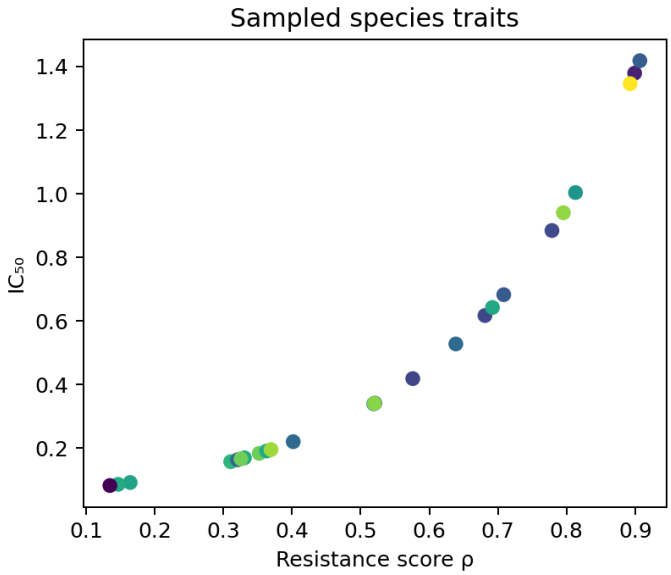
Resistance-placement experiment: the exact same 25 rho values and gLV community were retained while their species assignment was changed. Across 200 random assignments, treated/untreated final-biomass ratio had median 0.883 (5th–95th percentile 0.827–0.937). Assigning the highest rho values to highest-growth species gave ratio 0.854; assigning them to lowest-growth species gave 0.999. Because resistance cost is attached to rho, each assignment was compared to its own no-antibiotic control; this isolates antibiotic response from assignment-dependent baseline cost effects.

Results and figures

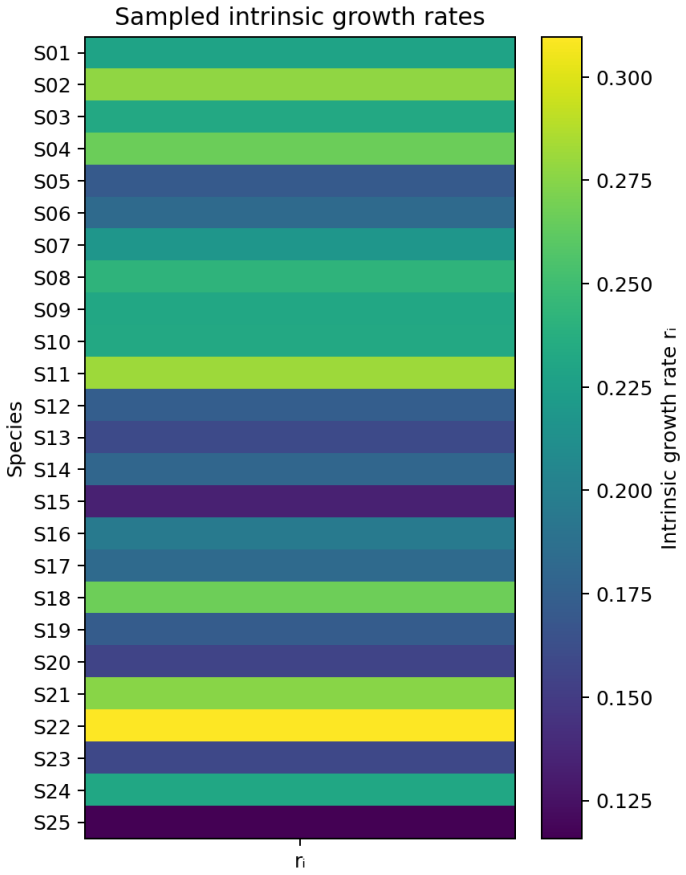
The results are exploratory model behavior, not fitted or validated biological estimates. A is held fixed within matched sweeps. The sensitivity analysis is a Spearman rank screen, not a variance-based global sensitivity analysis. Diagonal stability establishes a strong property for the autonomous gLV interaction system; it does not by itself prove stability of the full time-varying antibiotic system. All numerical conclusions should be read with the stated parameterization and finite simulation horizon in mind.

No-antibiotic reference and randomized community design.

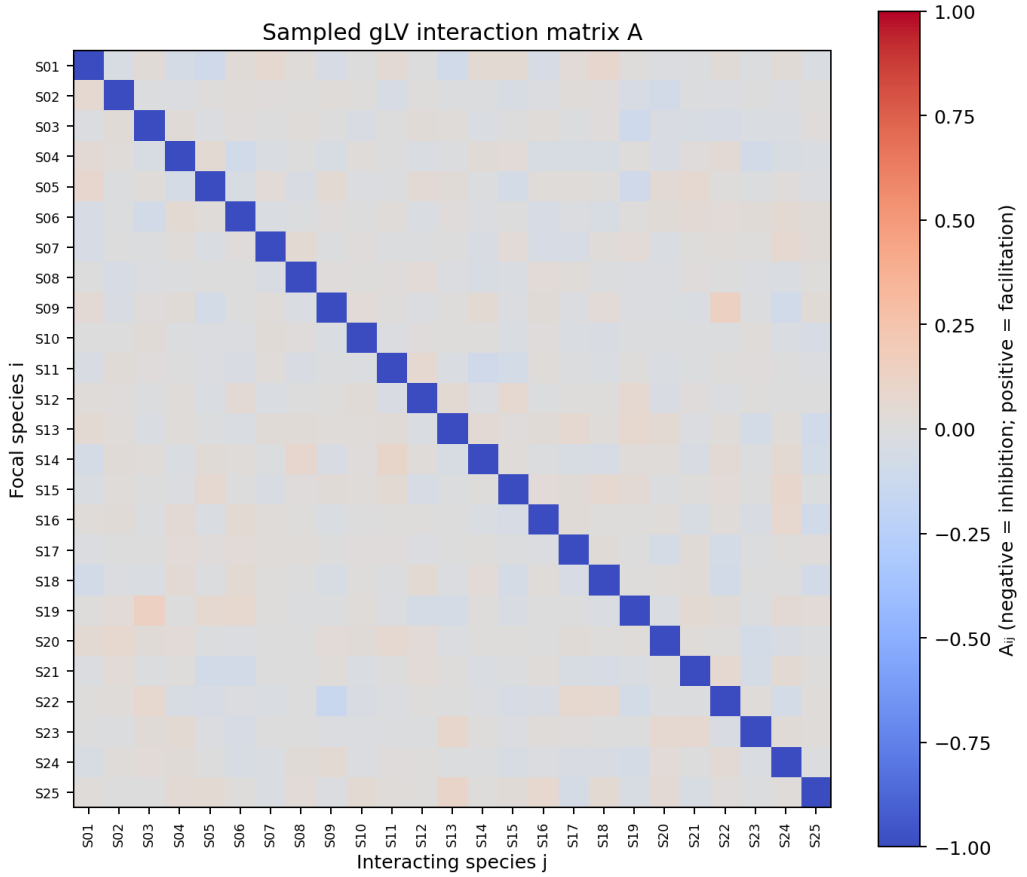




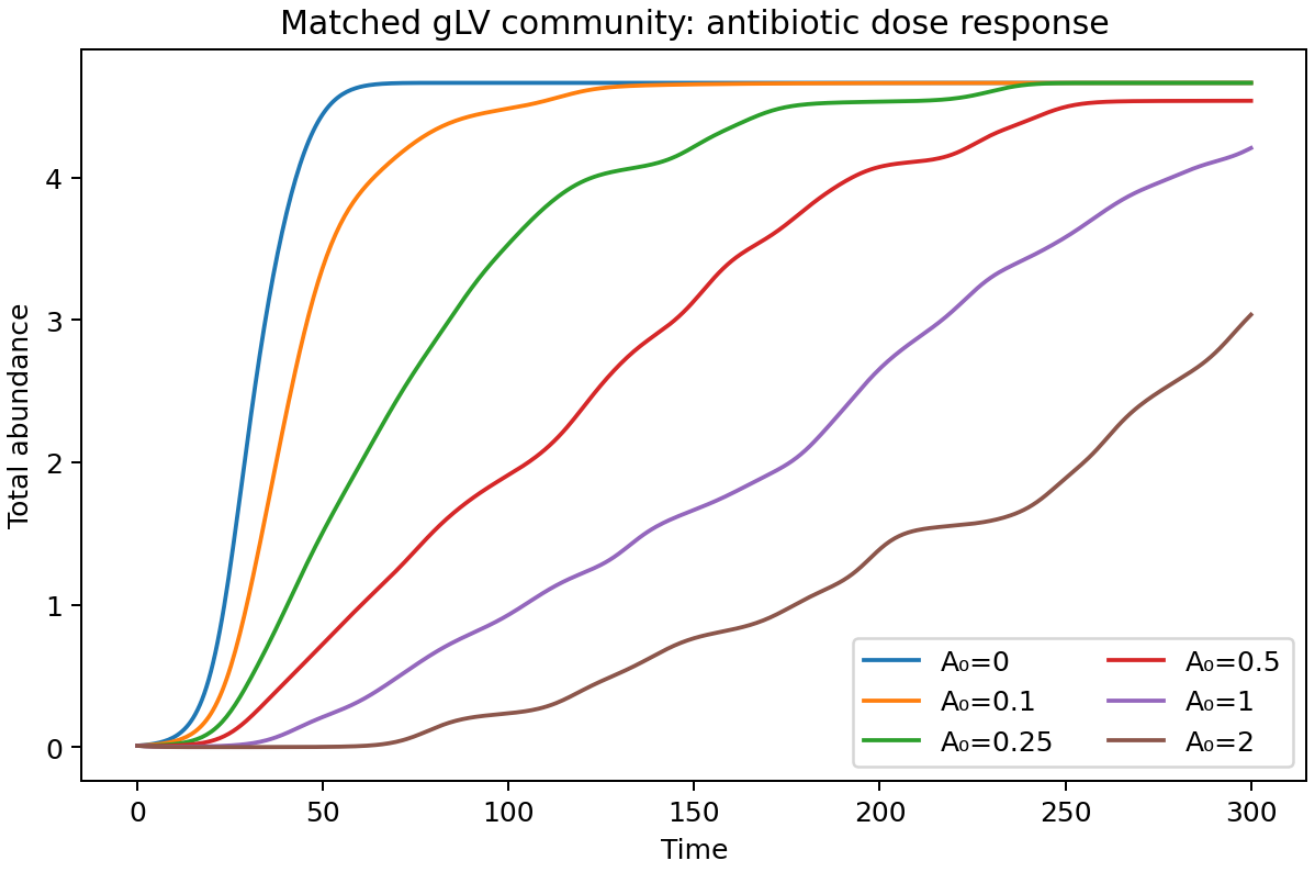
Sampled intrinsic growth-rate parameter for each species.



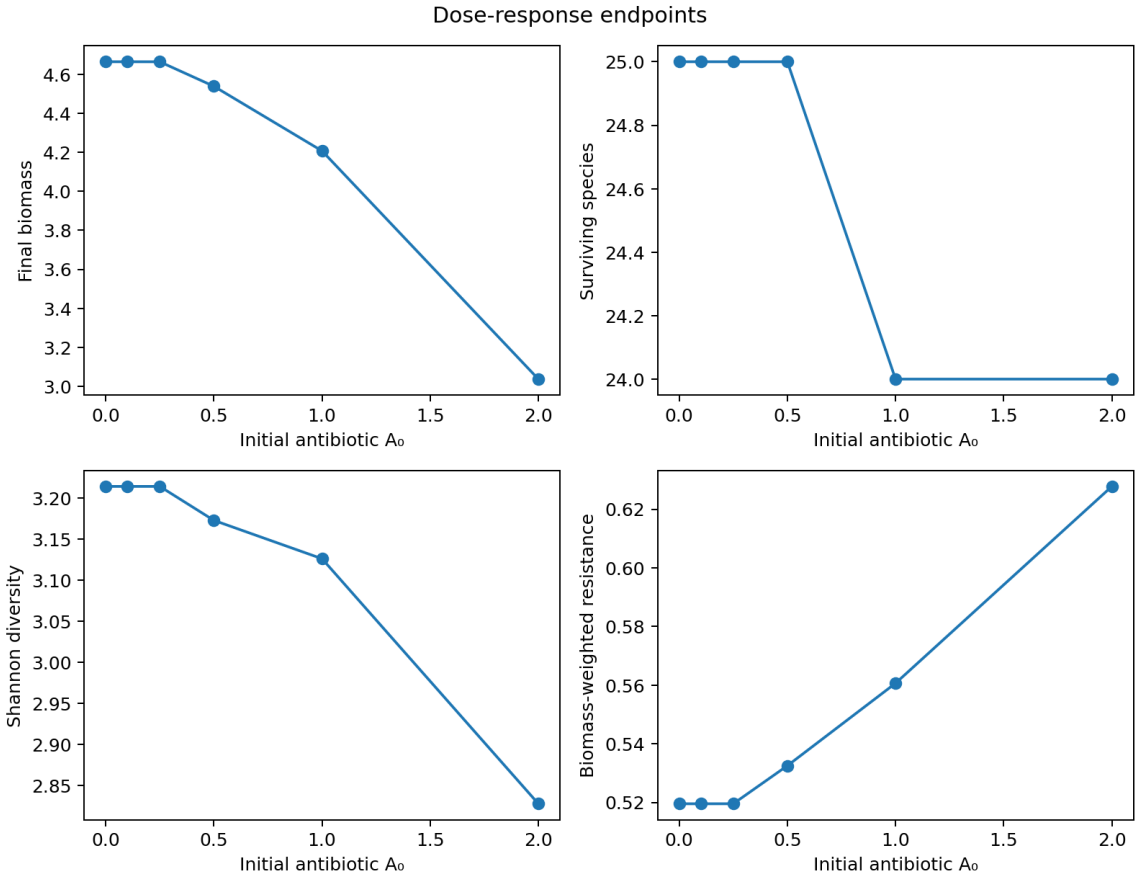
Sampled gLV interaction matrix; rows are focal species and columns are interacting species.



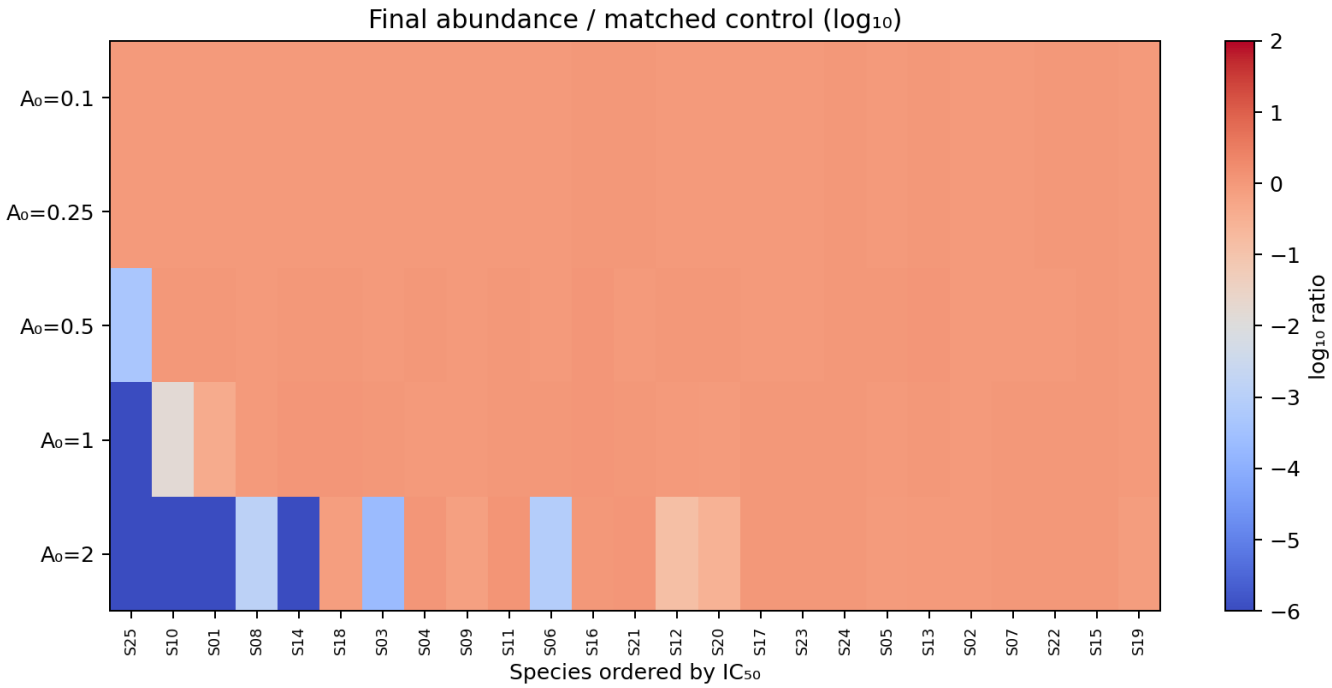
Dose response: total abundance, diversity, survivors, and resistance enrichment.



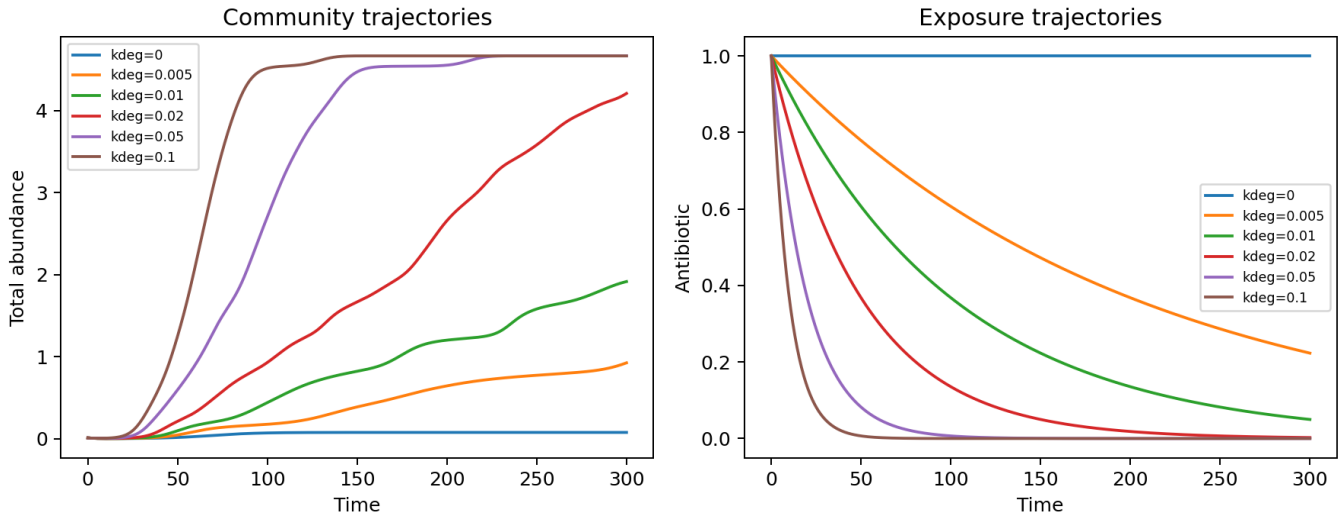
Dose-response endpoint metrics for abundance, survivors, diversity, and resistance.



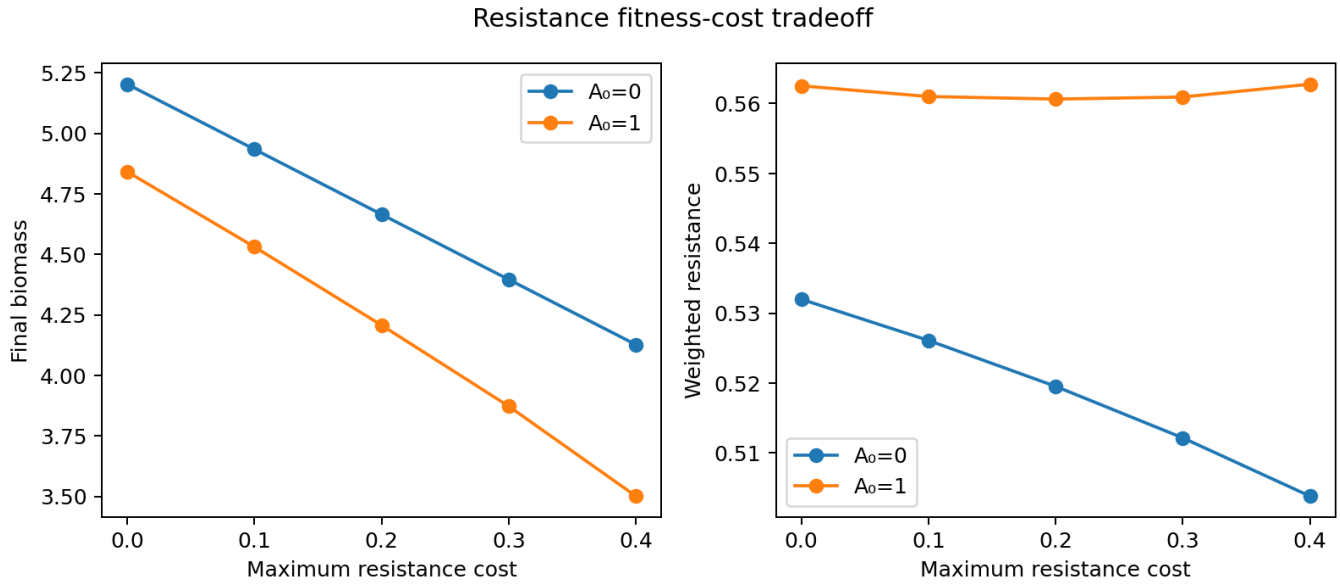
Species abundance ratios relative to matched no-antibiotic control.



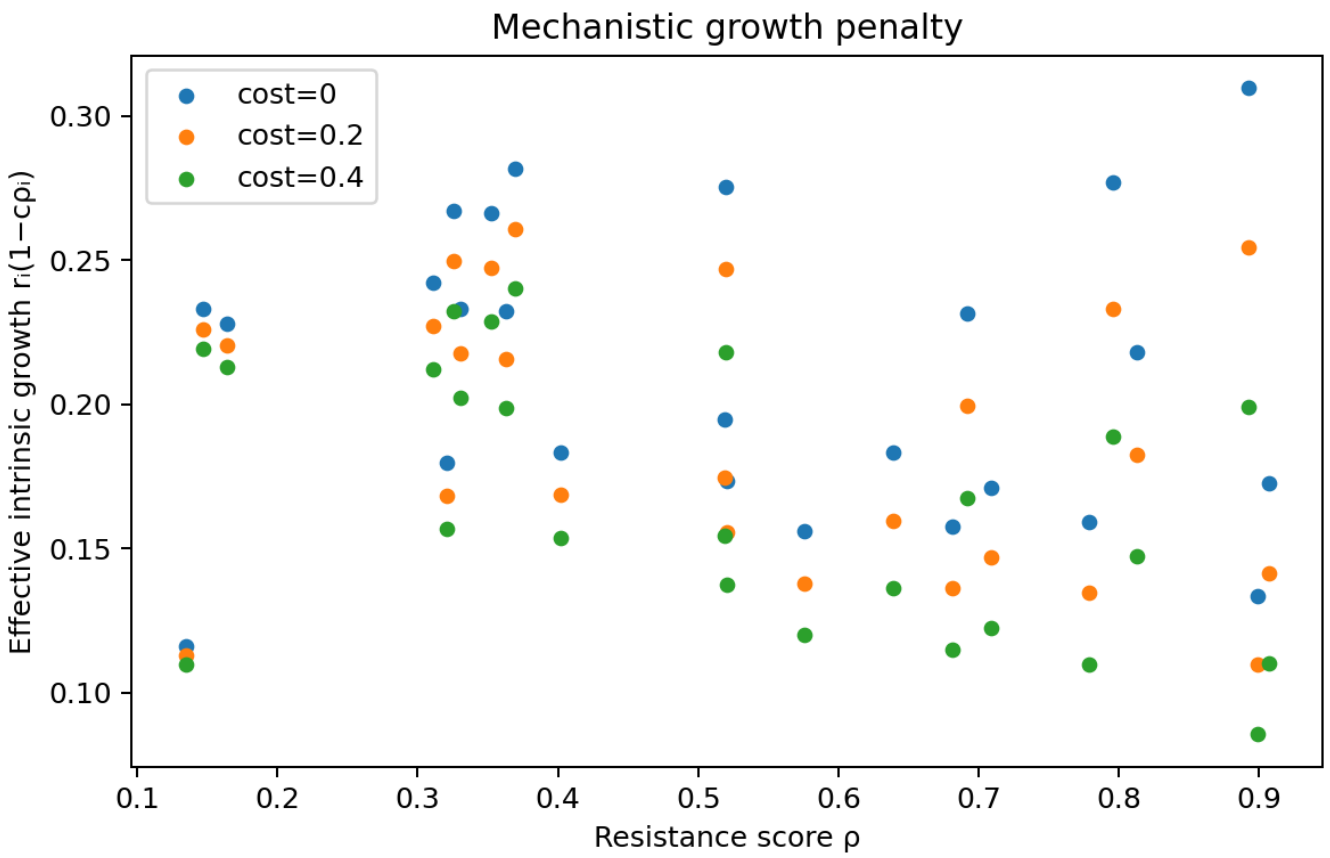
Drug persistence changes the duration of killing pressure.



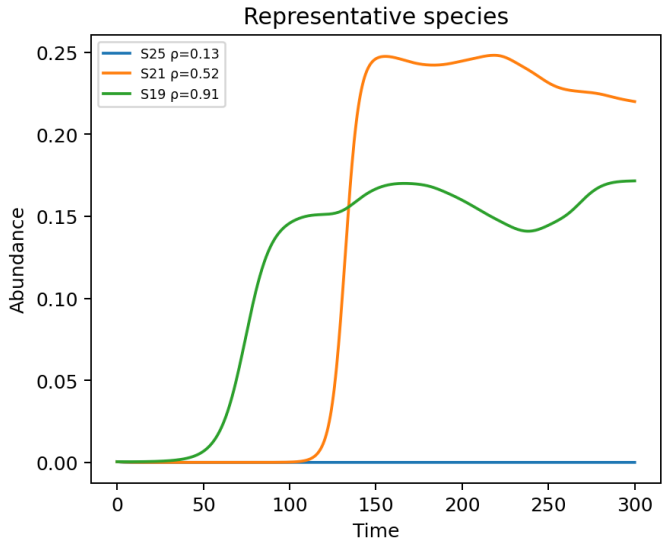
Resistance fitness-cost tradeoff under control and treatment.

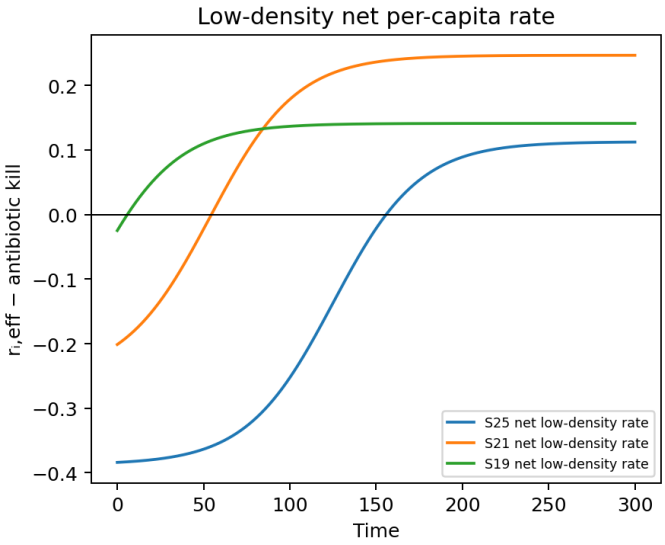


The explicit resistance-associated reduction in intrinsic growth.

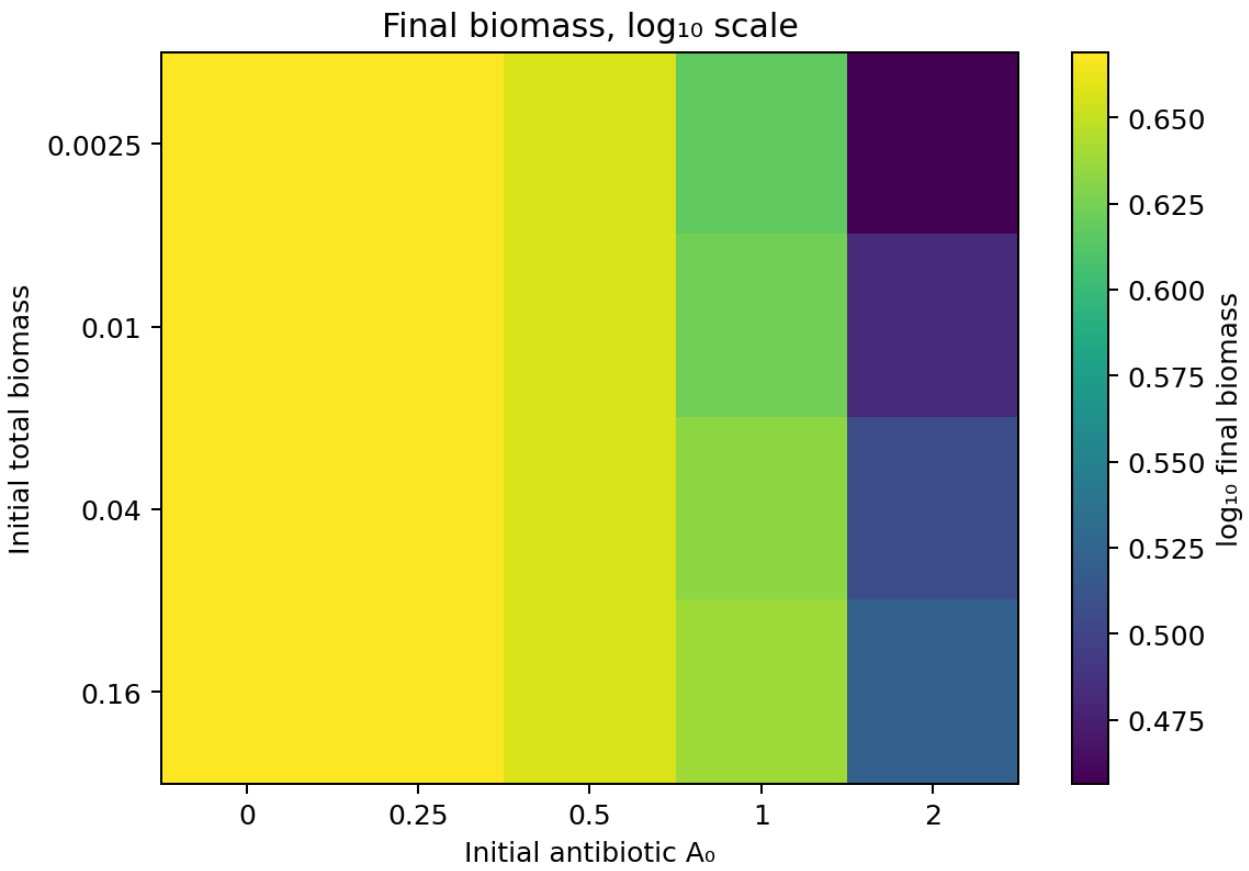


Representative intrinsic growth and direct antibiotic killing terms.

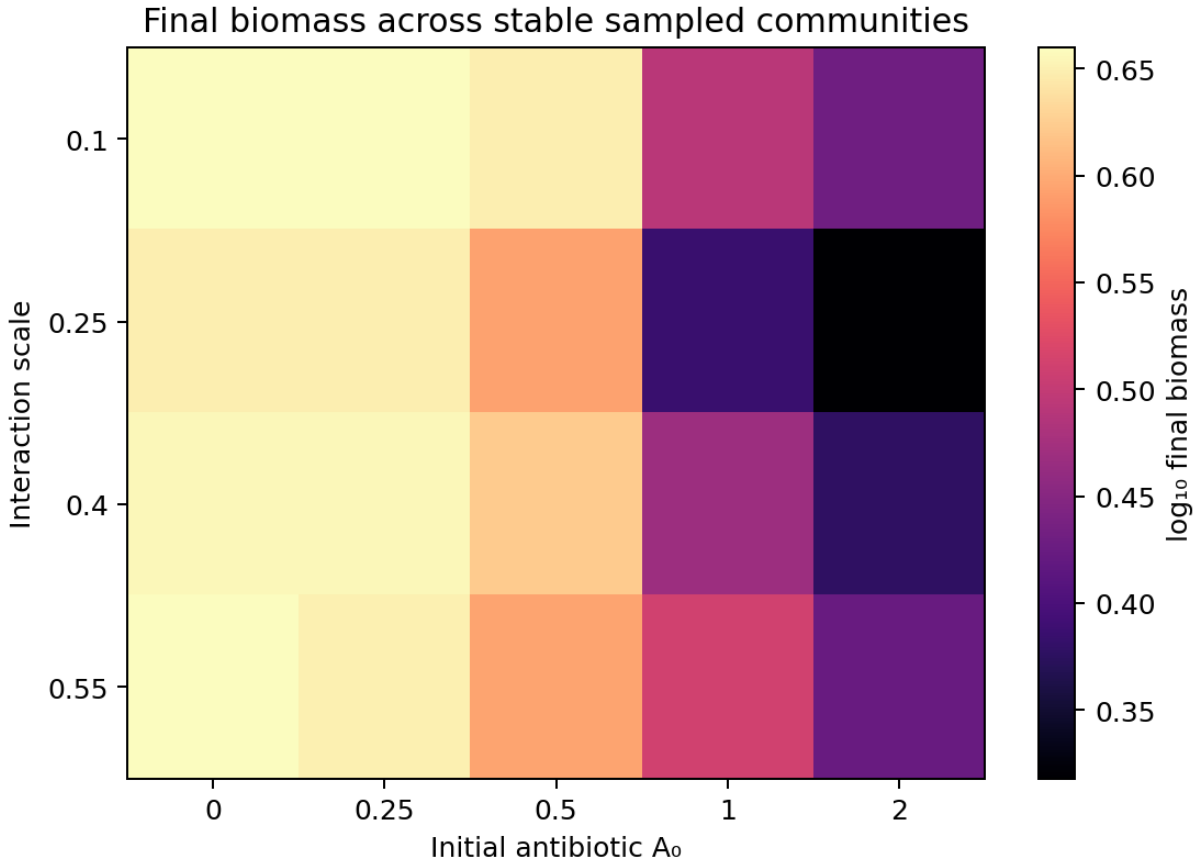




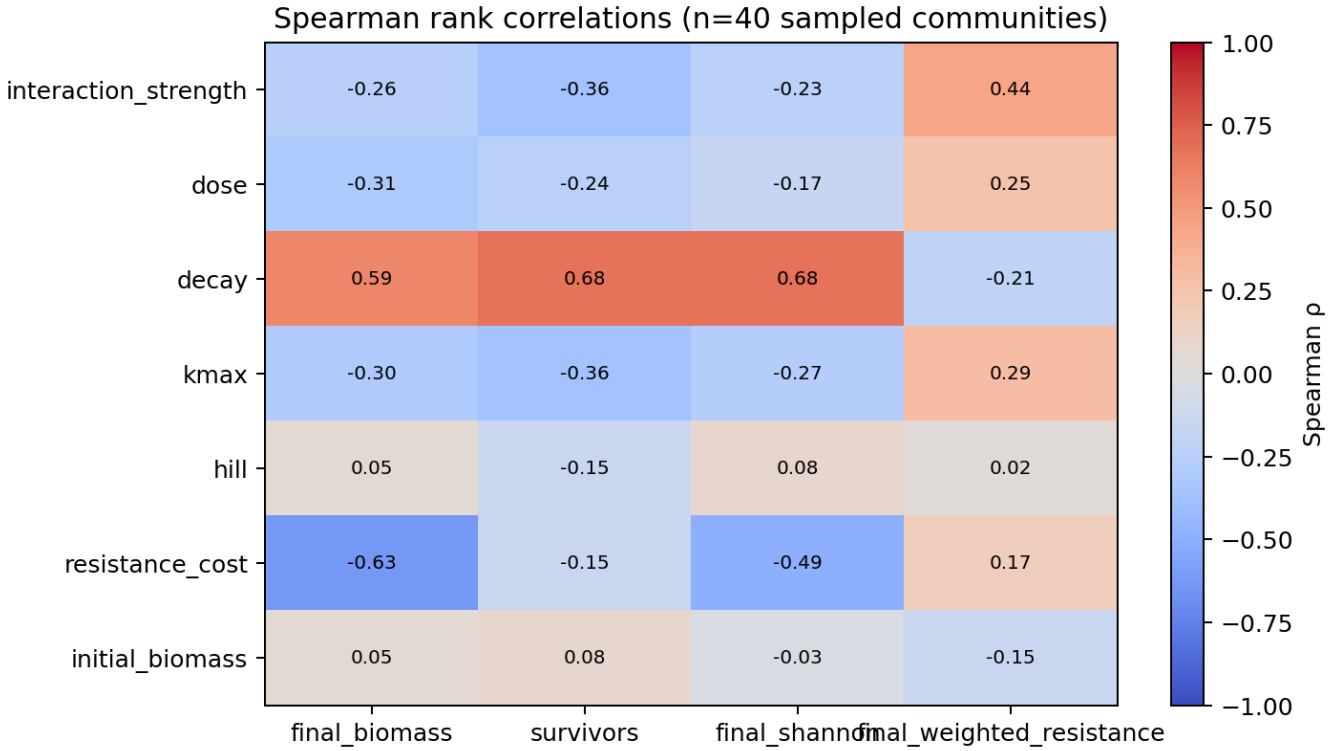
Initial community biomass-by-dose factorial.



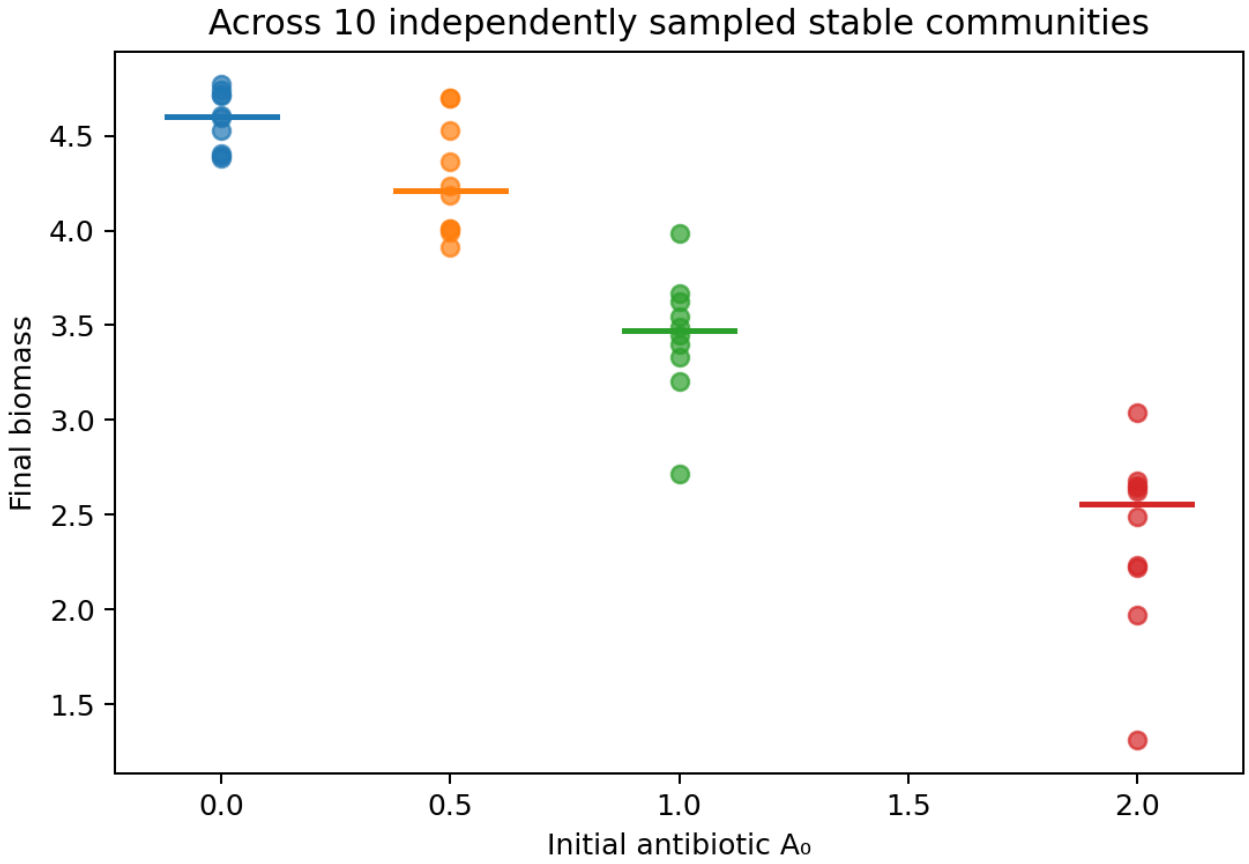
Interaction-strength-by-dose factorial across stable communities.



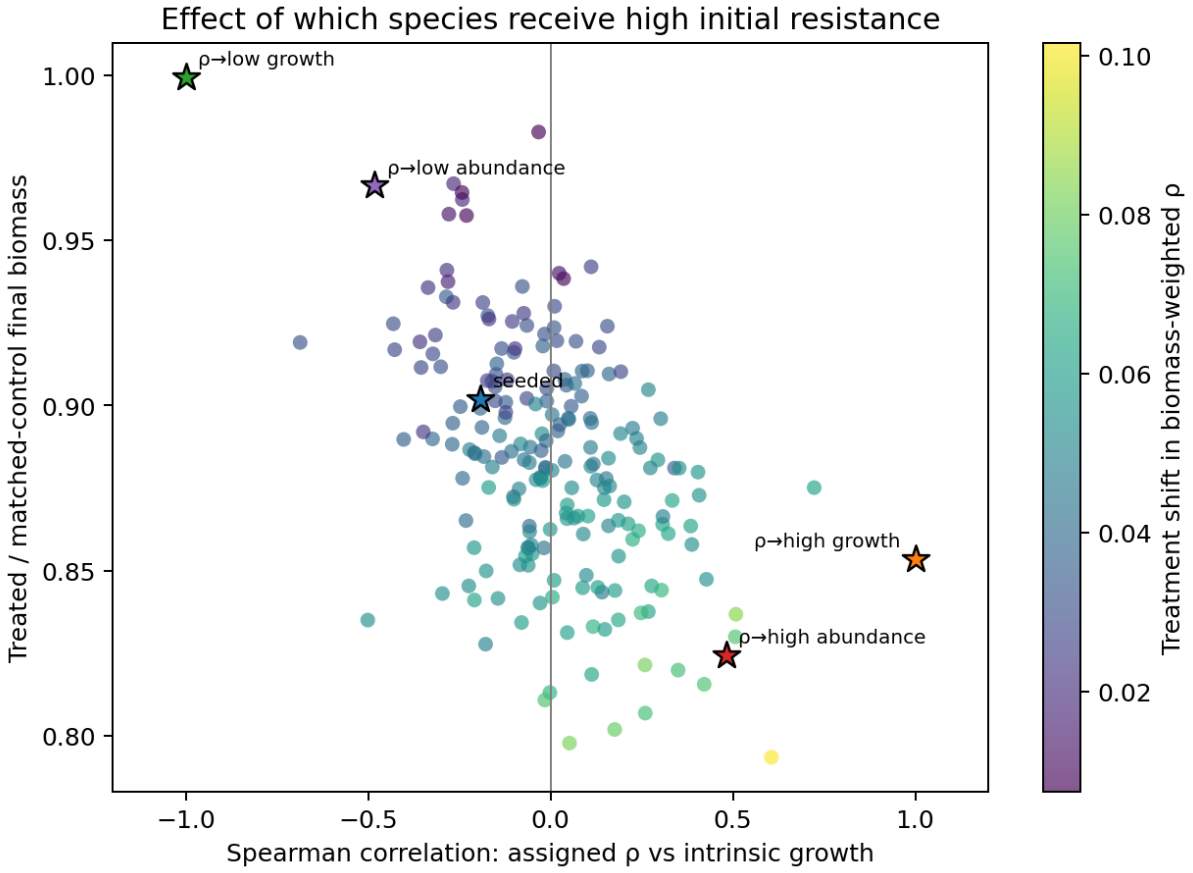
Rank correlations across randomly sampled stable communities and parameter ranges.



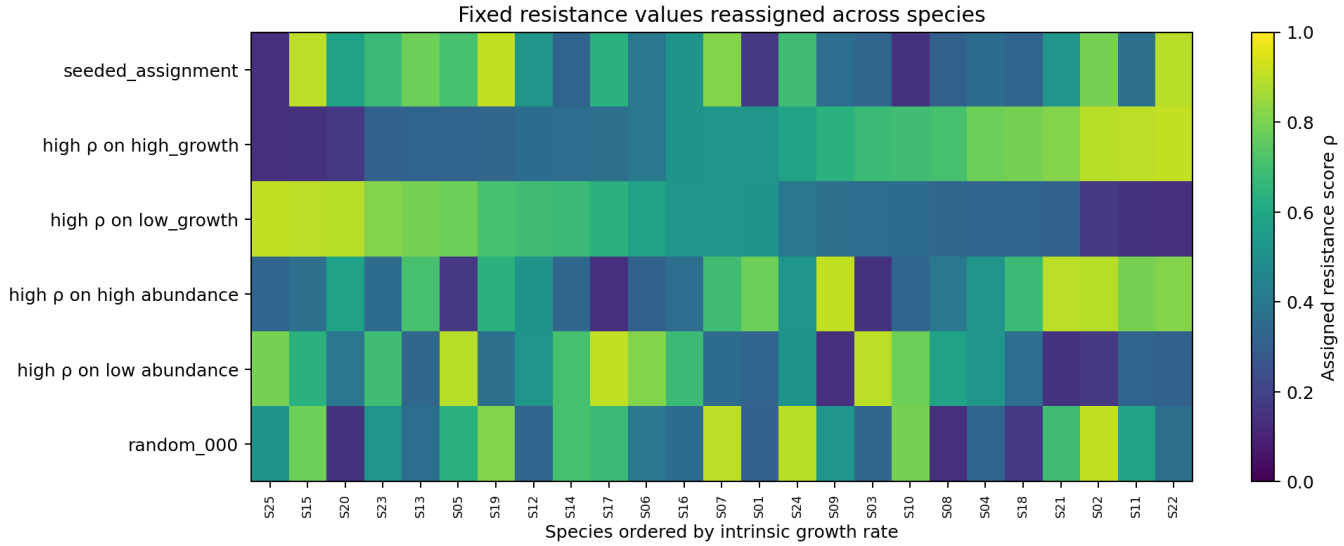
Variation across independently sampled community matrices.



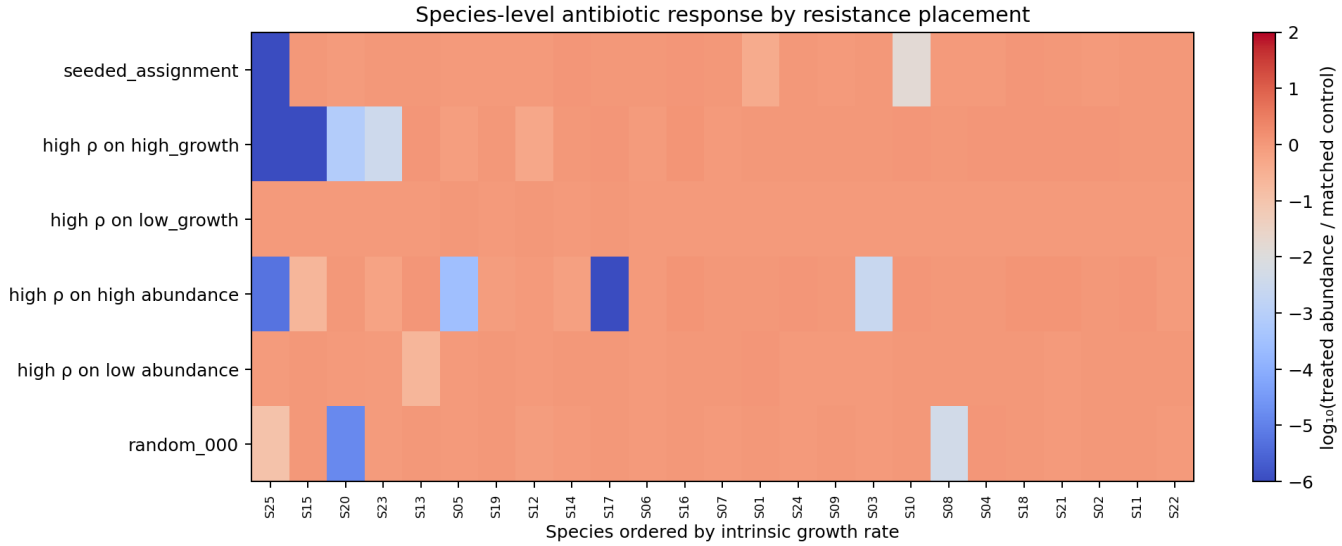
How rho-to-species assignment relates to antibiotic response while ecology and the rho multiset are held fixed.



The exact resistance values assigned to each species in the structured placements and example random assignment.



Matched treated/control species outcomes under distinct resistance placements.



How to read the main patterns

A decrease in total abundance after exposure reflects the balance between growth, density-dependent interactions, and antibiotic killing. A higher final biomass-weighted IC_{50} or resistance score than in the matched control is consistent with ecological selection for resistant species, but does not mean resistance arose during the simulation. Resistance is assigned at initialization. A cost can counter selection by reducing intrinsic growth. The biomass-by-dose factorial tests whether initial density changes the trajectory through gLV feedback; the interaction-strength factorial tests how the sampled community structure modifies treatment outcomes. Ensemble variability shows whether a result is specific to the fixed seed.

Supporting CSV files, full per-species outcomes, sampled matrices, model settings, and every plot in PNG/PDF formats are saved alongside this report in outputs/.